

Cam Follower Analysis

Problem Statement

F1_2: A cam follower with a mass of 4 kg moves in simple harmonic motion. It has a lift of 60 mm and a period of 0.2 seconds. Calculate the following:

- A. The velocity of the follower when it is 8 mm from the end of its travel.
- B. The maximum inertia force acting on the follower.
- C. The time taken for the follower to move from a point 15 mm before mid-travel to the end of its travel.

Given Data

- Mass of follower: $m = 4$ kg
- Lift (Amplitude): $A = 60$ mm = 0.06 m
- Period: $T = 0.2$ s

From the period, we calculate the angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.2} = 10\pi \text{ rad/s}$$

A: Velocity at 8 mm from End of Travel

The velocity in simple harmonic motion is given by:

$$v = \omega \sqrt{r^2 - x^2}$$

where:

- $\omega = \frac{2\pi}{T} = \frac{2\pi}{0.2} = 10\pi \text{ rad/s} = 31.42 \text{ rad/s}$
- $r = \frac{\text{Lift}}{2} = \frac{0.06}{2} = 0.03 \text{ m.}$
- $x = 30 - 8 = 22 \text{ mm.} = 0.022 \text{ m.}$

Now, the velocity:

$$v = 31.42 \sqrt{0.03^2 - 0.022^2} = \boxed{0.641 \text{ m/s}} \text{ 4 marks}$$

B: Maximum Inertia Force

The maximum inertia force is given by $F = m\alpha_{max}$ and $\alpha_{max} = \omega^2 r$.

The maximum inertia force:

$$F = 4 \times 31.42^2 \times 0.03 = \boxed{118.466 \text{ N}} \text{ 3 marks}$$

C: Time for Motion from 15 mm Before Mid-Travel to End

Given $x = 15 \text{ mm}$ and $r = 30 \text{ mm}$

$$\theta = \cos^{-1}\left(\frac{15}{30}\right) = 60^\circ$$

Thus, angle to travel through:

$$\alpha = 180^\circ - 60^\circ = 120^\circ$$

Since $\alpha = \omega t$,

$$t = \frac{\alpha}{\omega} = \frac{120 \times \pi}{180 \times 31.42} = \boxed{0.0667 \text{ s}} \text{ 3 marks}$$